Operating Systems

Sample educational document for testing a PDF question-answering chatbot.

1. Operating system

An operating system manages computer hardware and provides services to programs. Examples include process management, memory management, file management, and device control.

2. Process scheduling

CPU scheduling chooses which ready process runs next. First-Come, First-Served schedules in arrival order; Round Robin gives each process a time quantum; Shortest Job First selects a process with the smallest expected CPU burst.

3. Deadlock

A deadlock occurs when a set of processes wait indefinitely for resources held by one another. The four classic necessary conditions are mutual exclusion, hold and wait, no preemption, and circular wait.

4. Virtual memory

Virtual memory gives each process a logical address space and allows the operating system to use secondary storage as an extension of physical memory. Paging divides memory into fixed-size pages and frames.

5. Semaphore

A semaphore is a synchronization primitive used to coordinate concurrent processes or threads. Operations such as wait and signal can protect shared resources when used correctly.

Note: This is a short synthetic sample created for software testing, not an official textbook.